

RAM_POS	Opcode	Instruction	Output	Notes / Instruction Verifying
0	11110000	CLR A	A=0x00	Start state, clears carry (C=0)
1	0	LOAD 19, A	A<=M[19]	4-phase LOAD
2	10011	Page 0, 19	operand	LOAD operand byte
3	111000	BCD0 A	BCD0 "07"	Verifies BCD0 on A (A=0x07)
4	1000	OUT A	Out=0x07	Visible output
5	1	LOAD 20, B	B<=M[20]	4-phase LOAD
6	10100	Page 0, 20	operand	LOAD operand byte
7	111001	BCD0 B	BCD0 "03"	Verifies BCD0 on B (B=0x03)
8	1001	OUT B	Out=0x03	Visible output
9	10000000	ADD A	A=0x0A, C=0	0x07 + 0x03, no carry
10	111000	BCD0 A	BCD0 "10"	Verifies ADD result on display
11	1000	OUT A	Out=0x0A	Visible output
12	11110000	CLR A	A=0x00, C=0	Carry clear behavior
13	11110000	CLR A	A=0x00, C=0	Re-check CLR behavior
14	111000	BCD0 A	BCD0 "00"	Display clear value
15	1000	OUT A	Out=0x00	Visible output
16	110000	DEB 0, A	A=0x00/0x01	DEB result depends on debounced input
17	111000	BCD0 A	BCD0 "00"/"01"	Verifies DEB outcome
18	11110000	CLR A	A=0x00, C=0	Reset before carry tests
19	111	0x07 data	data	Referenced by LOAD 19,A
20	11	0x03 data	data	Referenced by LOAD 20,B
21	11010000	COM A	A=0xFF, C=1	COM must set carry
22	111000	BCD0 A	BCD0 "--"	0xFF non-BCD on both nibbles
23	1000	OUT A	Out=0xFF	Visible output
24	100100	BCS 31	branch taken	Carry is 1, BCS should branch
25	11111	Page 0, 31	operand	BCS target operand
26	11110000	CLR A	—	DOES NOT RUN (branch taken at 24)
27	111000	BCD0 A	—	DOES NOT RUN
28	1000	OUT A	—	DOES NOT RUN
29	100000	BCC 0	—	DOES NOT RUN
30	0	Page 0, 0	—	DOES NOT RUN
31	11110000	CLR A	A=0x00, C=0	CLR must clear carry
32	111000	BCD0 A	BCD0 "00"	Visible check after CLR
33	1000	OUT A	Out=0x00	Visible output
34	100000	BCC 40	branch taken	Carry is 0, BCC should branch
35	101000	Page 0, 40	operand	BCC target operand
36	11010000	COM A	—	DOES NOT RUN (branch taken at 34)
37	111000	BCD0 A	—	DOES NOT RUN
38	1000	OUT A	—	DOES NOT RUN
39	100000	BCC 0	—	DOES NOT RUN
40	0	LOAD 62, A	A<=M[62]	Load test data for carry ADD
41	111110	Page 0, 62	operand	LOAD operand byte
42	111000	BCD0 A	BCD0 "--"	A=0xFF shown
43	1000	OUT A	Out=0xFF	Visible output
44	1	LOAD 63, B	B<=M[63]	Load second ADD operand

45	111111	Page 0, 63	operand	LOAD operand byte
46	111001	BCD0 B	BCD0 "01"	B=0x01 shown
47	1001	OUT B	Out=0x01	Visible output
48	10000000	ADD A	A=0x00, C=1	0xFF + 0x01 overflow carry
49	111000	BCD0 A	BCD0 "00"	Result shown
50	1000	OUT A	Out=0x00	Visible output
51	100100	BCS 56	branch taken	Carry is 1, BCS should branch
52	111000	Page 0, 56	operand	BCS target operand
53	11110000	CLR A	—	DOES NOT RUN (branch taken at 51)
54	111000	BCD0 A	—	DOES NOT RUN
55	1000	OUT A	—	DOES NOT RUN
56	11110000	CLR A	A=0x00, C=0	Clear carry before loop
57	111000	BCD0 A	BCD0 "00"	Final visible state
58	1000	OUT A	Out=0x00	Final visible state
59	100000	BCC 0	branch taken	Infinite loop back to start
60	0	Page 0, 0	operand	BCC target operand
61	11110000	filler	—	Not reached in normal path
62	11111111	0xFF data	data	Reserved data for LOAD 62,A
63	1	0x01 data	data	Reserved data for LOAD 63,B